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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/093,962	01/06/2023	Yu Fang	SONA-0012-U01-C04	6795

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EXAMINER	
JHA, ABDHESH K	

ART UNIT	PAPER NUMBER
3668	

NOTIFICATION DATE	DELIVERY MODE
07/17/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/093,962	Applicant(s) Fang et al.	
	Examiner ABDHESH K JHA	Art Unit 3668	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 06/28/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 11-12 and 14-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 11-12 and 14-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 06/28/2024 and 12/18/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. Claims 1-19 are considered in this office action. Claims 1-9 are cancelled and new claims 10-19 are pending examination.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Arguments

3. Applicant's arguments filed 06/28/2024 have been fully considered but they are not persuasive. The applicant cancelled all the previous claims (1-10) and added new sets of claims 11-19.

Claim Objections

4. Claim 13 is objected as it seems to be a duplicate claim (Claim 12). Accordingly, the claim 13 is not been further treated on the merits.

Claim Rejections - 35 USC § 102

5. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

6. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

7. Claims 10-12, 14-16 are rejected under 35 U.S.C. 102(a)(1) based upon a public use or sale or other public availability of the invention Sugaiwa et. al (US 2018/0267530A1) and herein after Sugaiwa.

Regarding Claim 10, Sugaiwa teaches **a method** (Fig. 2 Para [0051]: "FIG. 2 is a flowchart illustrating one example of control routine concerning automated driving, which routine is executed by the automated driving system 100") comprising:

interpreting, via an automation definition circuit, an automation trigger description (Para [0053]: "Then, it is determined whether overwriting input is being entered from the user into the input device 9 (step S4).") **and an automation action description** (Para [0054]: "If an affirmative decision (YES) is obtained in step S4, the overwriting unit 14 executes overwriting operation to exclude the obstacle from the obstacle recognition result obtained by the obstacle recognizing unit 11 (step S5).");

providing, via an automation management circuit, a trigger detection plan and an automated action plan in response to the automation trigger description and the automation action description (Para [0054] "Namely, in step S5, the obstacle recognition result is overwritten into a result indicating that no obstacle was recognized. After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12;")

determining, via an automation execution circuit, a trigger event value in response to the trigger detection plan (Para [0055]: "In step S6, the traveling plan creating unit 12 creates a traveling plan, based on at least one of the detection results of the external sensors 3 and

the internal sensors 5, position information of the GPS receiver 4, map information of the map database 6, various kinds of information of the navigation system 7, and the obstacle recognition result")");

providing, via the automation execution circuit, an automation command in response to the trigger event value and the automated action plan; and implementing, in response to the automated action plan via an end point of a plurality of end points of a vehicle having a network, an automated vehicle response (Para [0055] : "Then, the vehicle controller 13 performs vehicle control based on the traveling plan thus created, and causes the vehicle V to autonomously travel according to the traveling plan.").

Regarding Claim 11, Sugaiwa teaches the method of claim 10. Sugaiwa also teaches the **automation trigger description comprises a filter description for a vehicle operating parameter, and providing, via the automation management circuit, the trigger detection plan and the automated action plan is further in response to the filter description for the vehicle operating parameter** (Para [0080-0083] : "The reliability of the obstacle recognition result obtained by the obstacle recognizing unit 11 is calculated by the reliability calculating unit 15 (step S24). As in the above step S4, it is determined whether the user is entering overwriting input into the input device 9 (step S25). [0081] If an affirmative decision (YES) is obtained in step S25, it is determined whether the calculated reliability is smaller than the threshold value (step S26). If an affirmative decision (YES) is obtained in step S26, the overwriting unit 14 performs overwriting operation (step S27). After execution of step S27, the obstacle recognition result subjected to the overwriting operation is generated to the traveling plan creating unit 12, and the control proceeds to step S28 that will be described later. [0082] On the other hand, if a

negative decision (NO) is obtained in step S25, or a negative decision (NO) is obtained in step S26, the overwriting unit 14 does not perform the overwriting operation, and the obstacle recognition result obtained by the obstacle recognizing unit 11 is generated as it is to the traveling plan creating unit 12. Then, the control proceeds to step S28 that will be described later. In step S28, the traveling plan creating unit 12 creates a traveling plan, as in the above step S6. Then, as in the above step S7, the vehicle controller 13 performs vehicle control based on the traveling plan, so that the vehicle V autonomously travels according to the traveling plan (step S29).").

Regarding Claim 12, Sugaiwa teaches the method of claim 10. ***Sugaiwa also teaches*** **the automation trigger description comprises a data parsing description for at least one of a vehicle operating parameter, a network message, or a controller parameter** (Para [0053]: "If a negative decision (NO) is obtained in step S2, the control proceeds to step S6 that will be described later. If an affirmative decision (YES) is obtained in step S2, the obstacle recognizing unit 11 recognizes the obstacle based on the detection result of the external sensor(s) 3 (step S3). Then, it is determined whether overwriting input is being entered from the user into the input device 9 (step S4).").;

and providing, via the automation management circuit, the trigger detection plan and the automated action plan is further in response to the data parsing description (Para [0054]: "If an affirmative decision (YES) is obtained in step S4, the overwriting unit 14 executes overwriting operation to exclude the obstacle from the obstacle recognition result obtained by the obstacle recognizing unit 11 (step S5). Namely, in step S5, the obstacle recognition result is

overwritten into a result indicating that no obstacle was recognized. After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12.").

Regarding Claim 14, Sugaiwa teaches the method of claim 10. ***Sugaiwa also teaches*** **the automation trigger description comprises a map description** (Para [0037]: "the map database 6 is a database that stores map information. The map database 6 is formed in an HDD (Hard Disk Drive) installed on the vehicle V. The map information includes position information of roads, lane information, road type information, road shape information, position information of intersections and branching points, position information of buildings, and so forth. The road type information is used for discriminating the types of roads, such as a limited highway (or expressway), and a general road. The road shape information includes type information, such as a curved section, and a straight section, and the curvature of the road, for example. The map database 6 may be stored in a computer of a facility, such as an information processing center, which can communicate with the vehicle V.");

and providing, via the automation management circuit, the trigger detection plan and the automated action plan is further in response to the map description ([0038] "The navigation system 7 is a system that guides the user of the vehicle V to a preset destination. The navigation system 7 recognizes a traveling road and a traveling lane on which the vehicle V is traveling, based on the position of the vehicle V measured in the GPS receiver 4, and the map information of the map database 6. The navigation system 7 computes a target route from the position of the vehicle V to the destination. The navigation system 7 guides the user along the target route, using

a display panel and a speaker. The navigation system 7 transmits the position information of the vehicle V, information on the traveling lane of the vehicle V, and information on the target route of the vehicle V, to the ECU 10.").

Regarding Claim 15, Sugaiwa teaches the method of claim 14. ***Sugaiwa also teaches*** wherein the map description comprises a multi-parameter value map for a plurality of values, the plurality of values each comprising at least one of: a vehicle operating parameter, an external operating parameter, a fault value, or a state value (Para [0044]: "The traveling plan creating unit **12** creates the long-term traveling plan of the vehicle V, based on the target route set by the navigation system **7** and the map information of the map database **6**. The long-term traveling plan has a control target value of the vehicle V corresponding to a position on the target route of the vehicle V. The position on the target route is a position in a direction in which the target route extends on the map. The position on the target route means a set vertical position set for each given interval (e.g., 1 m) in the direction of extension of the target route. The control target value provides a control target of the vehicle V in the long-term traveling plan. The control target value is set in association with each of the set vertical positions on the target route. The traveling plan creating unit **12** sets the set vertical positions at the given intervals on the target route, and also sets a control target value for each of the set vertical positions, to thus create the long-term traveling plan. The set vertical position and a target lateral position may be combined and set as one position coordinate. The set vertical position and target lateral position mean information of a vertical position and information of a lateral position set as targets in the long-term traveling plan.").

Regarding Claim 16, Sugaiwa teaches the method of claim 14. ***Sugaiwa also teaches*** **the automation trigger description comprises an aggregated data description; and providing, via the automation management circuit, the trigger detection plan and the automated action plan is further in response to the aggregated data description** (Para [0055] : "In step S6, the traveling plan creating unit 12 creates a traveling plan, based on at least one of the detection results of the external sensors 3 and the internal sensors 5, position information of the GPS receiver 4, map information of the map database 6, various kinds of information of the navigation system 7, and the obstacle recognition result. Then, the vehicle controller 13 performs vehicle control based on the traveling plan thus created, and causes the vehicle V to autonomously travel according to the traveling plan.")

Claim Rejections - 35 USC § 103

8. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

9. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

10. The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

11. Claims 17-19 are rejected under 35 U.S.C. 103 as being unpatentable over Sugaiwa in view of Konrardy et al. (US2021/0116256) and herein after will be referred as Konrardy.

Regarding Claim 17, Sugaiwa teaches the method of claim 14.

Konrardy teaches the automation trigger description comprises a statistical analysis description; and providing, via the automation management circuit, the trigger detection plan and the automated action plan is further in response to the statistical analysis description (Fig 7 Step 702-718).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified Sugaiwa to incorporate the teachings of Konrardy to include statistical analysis. Doing so would optimize the system to implement the plan accordingly.

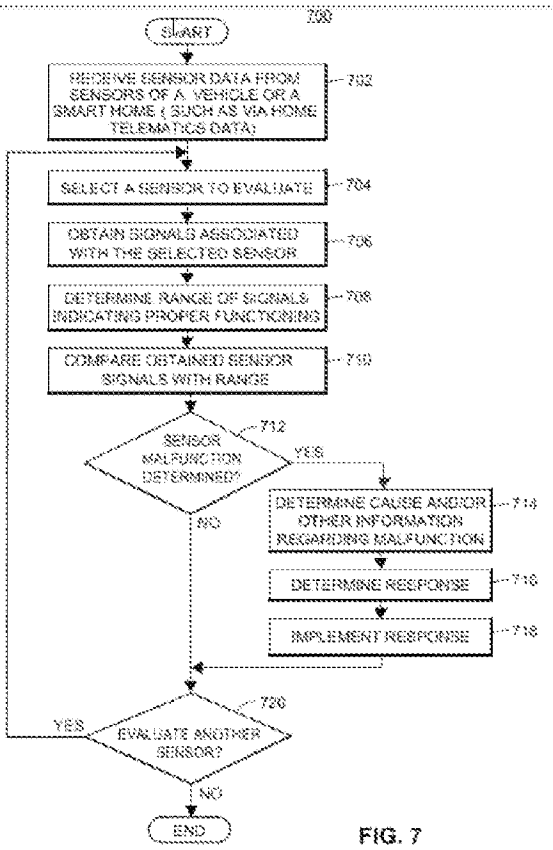


FIG. 7

Regarding Claim 18, Sugaiwa in view of Konrardy teaches the method of claim 17. Konrardy teaches wherein the statistical analysis description comprises a statistical anomaly definition for a vehicle operating parameter (Fig.7 #712-714).

Regarding Claim 19, Sugaiwa in view of Konrardy teaches the method of claim 18. Konrardy teaches determining, via the automation execution circuit and in response to the statistical anomaly definition and values of the vehicle operating parameter, that a relevant operating condition is present; and determining, via the automation execution circuit in response to the relevant operating condition, a trigger event value (Fig.7).

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ABDHESH K JHA whose telephone number is (571)272-6218. The examiner can normally be reached M-F:0800-1700.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, James J Lee can be reached on 571-270-5965. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ABDHESH K JHA/
Primary Examiner, Art Unit 3668